

Data Design for Engineering Fun and Profit

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How Long Have Computers Been in Structural Engineering?

1990s?

1980s?

1970s?

...1960s?

Here is what people were doing in 1966

"We have not created the software yet..."

A CAD research program at the
University of London Atlas Computing
Service



<https://youtu.be/oRBS70J2Poo?si=Yo3OWa0kr7qp5ByY&t=995>

What you just saw

"We have not created the software yet..."

This CAD program would have been research software developed at the University of London Atlas Computing Service and was directly inspired by Sketchpad, developed at MIT by Ivan Sutherland in 1963

The ideas for the software we are currently using were created over 60 years ago

We have just got slightly fancier versions of the same ideas

'No no...I do engineering. I don't process data.'

I am arguing that "engineering" can be broken into five general steps

1. Gathering information
2. Performing a simulation
3. Making decisions
4. Validating those decisions
5. Reporting

Each of those steps requires data to be processed, either by us or by a computer

Until we understand data, we will never be in control

I am here to share what I have learned about data

- How to recognize different structures
 - Tables
 - Trees
- Different representation formats for tables and trees
 - Strengths and weaknesses
 - Why you might use one over the other

Tables and Trees

Tables: a collection of records (rows) each with the same fields (columns)

Trees: a hierarchy of nodes where each node holds a value, an array of values, or another tree

Tree values

A value can be a scalar, an array, or another tree

```
{  
  node1: value1,  
  node2: [v2.1, v2.2],  
  node3: {  
    node4: value4  
  }  
}
```

Comparing Trees and Tables

Finite element models, an example everyone is familiar with

Commonly (but not always):

- Finite element analysis models are created as trees
- Finite element analysis results are created as tables

Which is a mistake:

- Finite element analysis models should be tables
- Finite element analysis results should be trees